

Deliverables

Commit and push all your changes to your remote repository.

Your repository must include the following:

- The design directory (Do not commit the design with the bug fix)
- The test directory
- The sim directory containing lab_report.pdf (Format is already present in the sim directory)
- The tb directory containing updated files.

Important note: To get full credit, you must upload all the required files and directories and strictly name your files according to the requirements.

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Course:Lab-10-616-Report:

Coverage Report :

Failing Regression:

The screenshot shows the vManager software interface with a table of regression sessions. The table has columns for Session Status, Name, Total Runs, #Passed, #Failed, #Running, #Waiting, #Other, Start Time, and Owner. The sessions are listed with their respective run counts and status indicators (green for passed, red for failed).

Session Status	Name	Total Runs	#Passed	#Failed	#Running	#Waiting	#Other	Start Time	Owner
completed	htax_regress.shes123.25_12_04_15_13_48_6609	60	57	3	0	0	0	12/4/25 3:13 PM	shes123
completed	htax_regress.shes123.25_12_04_15_07_21_3100	10	10	0	0	0	0	12/4/25 3:07 PM	shes123
completed	htax_regress.shes123.25_12_03_21_43_36_3203	15	0	15	0	0	0	12/3/25 9:43 PM	shes123
completed	htax_regress.shes123.25_12_03_21_36_27_6950	15	0	15	0	0	0	12/3/25 9:36 PM	shes123
completed	htax_regress.shes123.25_11_19_18_36_18_3695	18	10	8	0	0	0	11/19/25 6:36 PM	shes123
completed	htax_regress.shes123.25_11_19_18_09_03_4246	18	10	8	0	0	0	11/19/25 6:09 PM	shes123
completed	htax_regress.shes123.25_11_19_17_38_54_0924	18	10	8	0	0	0	11/19/25 5:38 PM	shes123

Failing Regression Passing:

Session Status	Name	Total Runs	#Passed	#Failed	#Running	#Waiting	#Other	Start Time	Owner
(no filter)	(no filter)	(no filter)	(no filter)	(no filter)	(no filter)	(no filter)	(no filter)	(no filter)	==SME()
post_session_script	htax_regress.shes123.25.12.04.15.35.18.9587	60	60	0	0	0	0	12/4/25 3:35 PM	shes123
completed	htax_regress.shes123.25.12.04.15.13.48.6609	60	57	3	0	0	0	12/4/25 3:13 PM	shes123
completed	htax_regress.shes123.25.12.04.15.07.21.3100	10	10	0	0	0	0	12/4/25 3:07 PM	shes123
completed	htax_regress.shes123.25.12.03.21.43.36.3203	15	0	15	0	0	0	12/3/25 9:43 PM	shes123
completed	htax_regress.shes123.25.12.03.21.36.27.6950	15	0	15	0	0	0	12/3/25 9:36 PM	shes123
completed	htax_regress.shes123.25.11.19.18.36.18.3695	18	10	8	0	0	0	11/19/25 6:36 PM	shes123
completed	htax_regress.shes123.25.11.19.18.09.03.4246	18	10	8	0	0	0	11/19/25 6:09 PM	shes123
completed	htax_regress.shes123.25.11.19.17.38.54.0924	18	10	8	0	0	0	11/19/25 5:38 PM	shes123

Tx mapping tests:

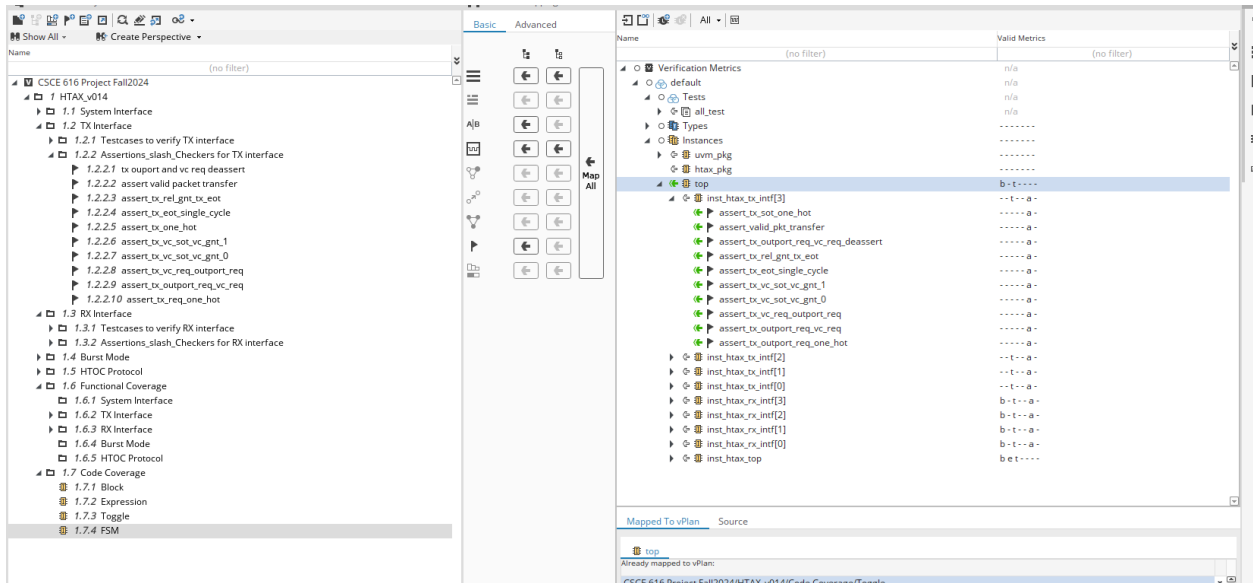
CSCE 616 Project Fall2024

- 1 HTAX_v014
 - 1.1 System Interface
 - 1.2 TX Interface
 - 1.2.1 Testcases to verify TX interface
 - 1.2.1.1 Simple Port-Port test
 - 1.2.1.2 Short Packet test
 - 1.2.1.3 Random test
 - 1.2.1.4 Multiport sequential_test
 - 1.2.1.5 Medium Packet Fixed
 - 1.2.1.6 Long Packrt Fixed
 - 1.2.1.7 Fixed Length Fixed Delay
 - 1.2.2 Assertions_slash_Checkers for TX interface
 - 1.3 RX Interface
 - 1.3.1 Testcases to verify RX interface
 - 1.3.2 Assertions_slash_Checkers for RX interface
 - 1.4 Burst Mode
 - 1.5 HTOC Protocol
 - 1.6 Functional Coverage
 - 1.6.1 System Interface
 - 1.6.2 TX Interface
 - 1.6.3 RX Interface
 - 1.6.4 Burst Mode
 - 1.6.5 HTOC Protocol
 - 1.7 Code Coverage
 - 1.7.1 Block
 - 1.7.2 Expression
 - 1.7.3 Toggle
 - 1.7.4 FSM

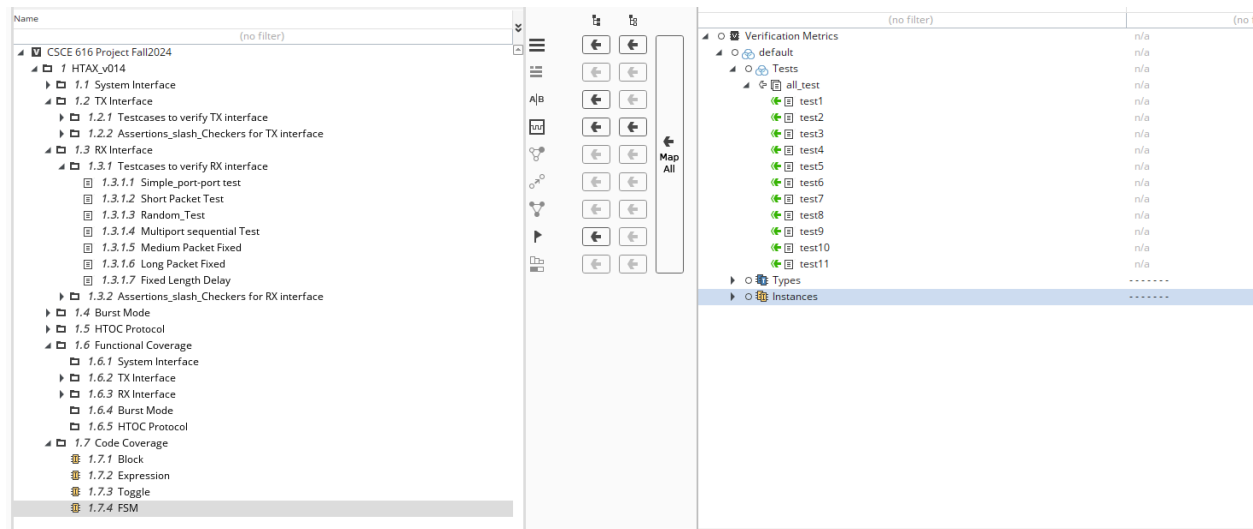
Verification Metrics

- default
 - Tests
 - all_test
 - test1
 - test2
 - test3
 - test4
 - test5
 - test6
 - test7
 - test8
 - test9
 - test10
 - test11
 - Types
 - Instances
 - uvm_pkg
 - htax_pkg
 - top

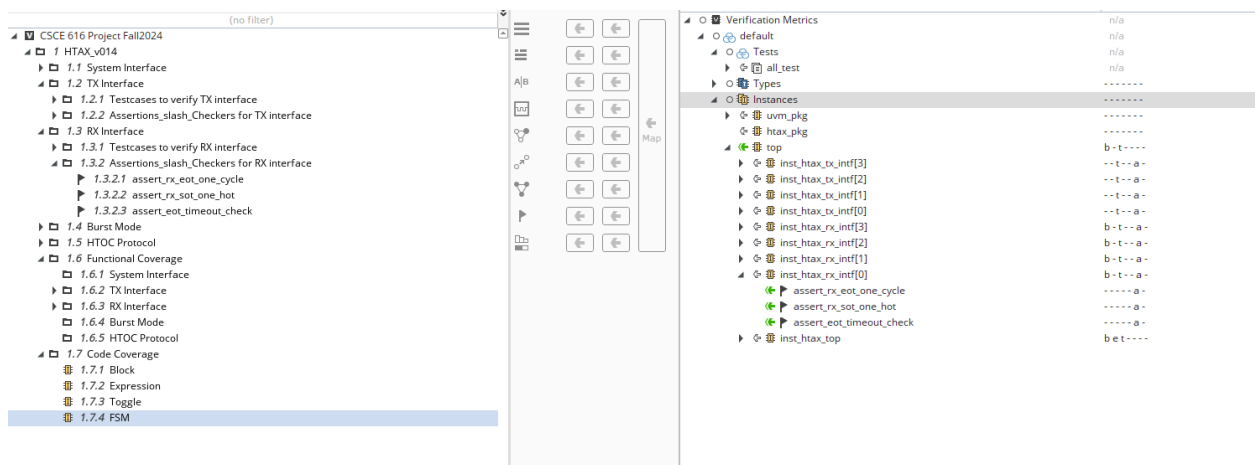
Tx Assertions Mapping:



Rx tests mapping:

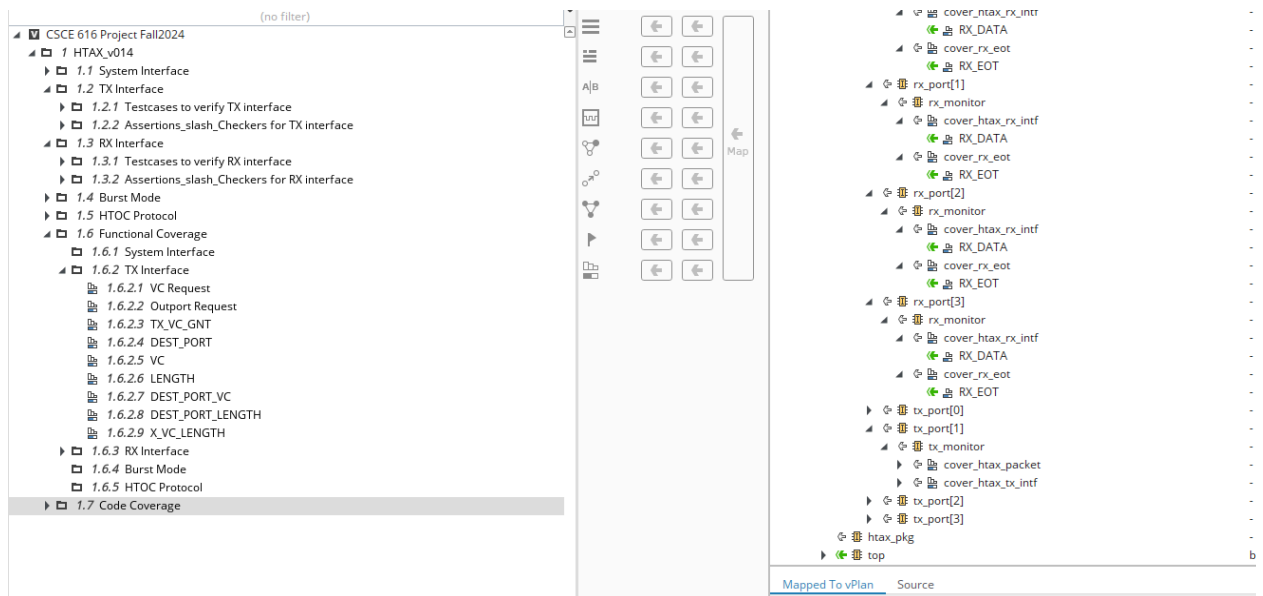


Rx Assertions Mapping:

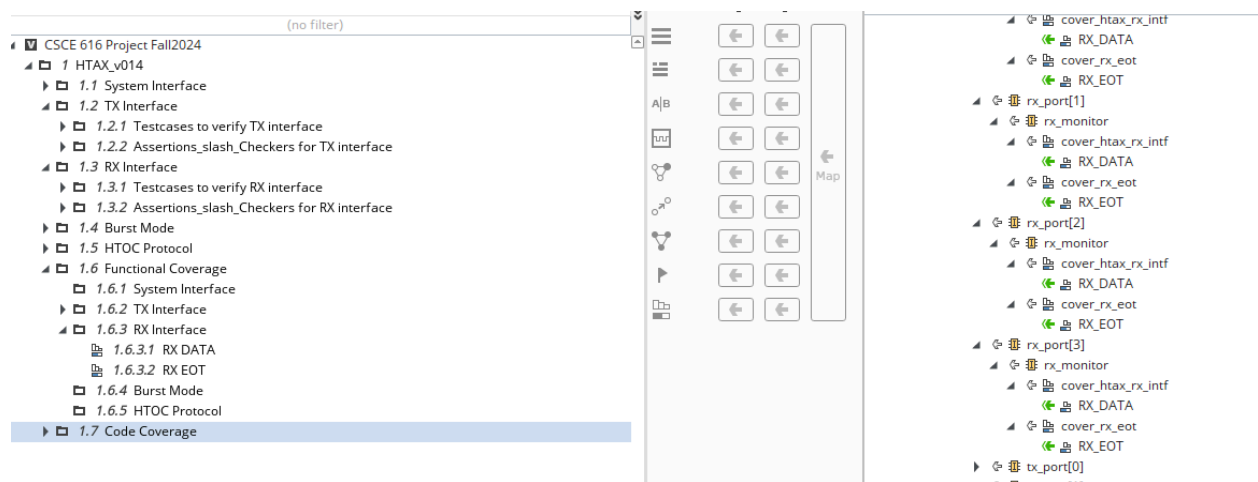


Functional Coverage Mapping:

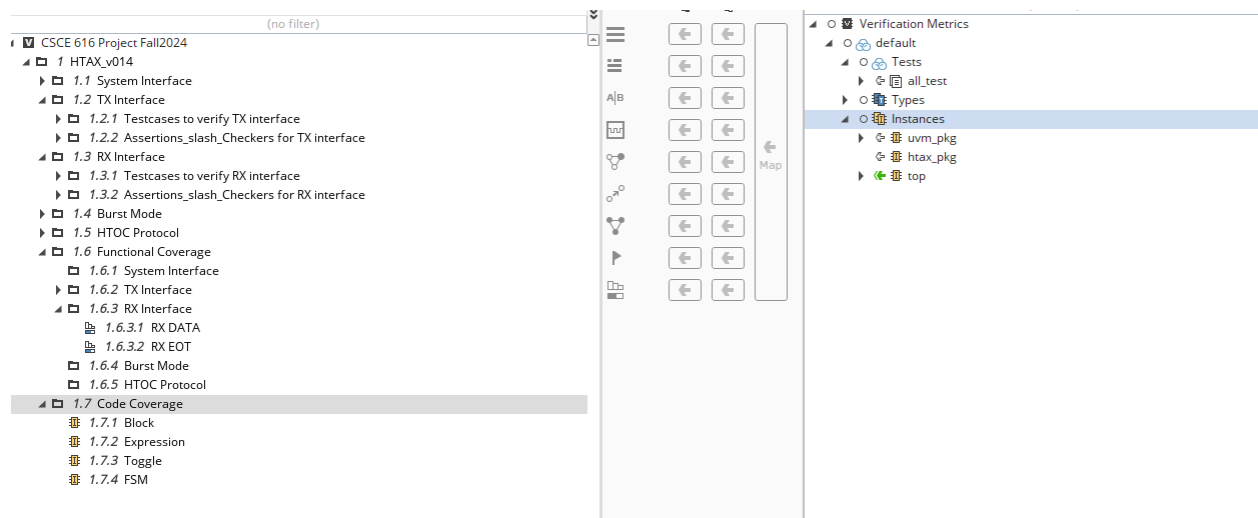
Tx functional coverage:



Rx functional coverage:



Code Coverage Mapping:



Analysis coverage results:

1.tx interface tests:

Ex	URI	Name	Overall Average Grade	Overall Covered	Assertion Status Grade
		(no filter)	(no filter)	(no filter)	(no filter)
		1.2.1.1 Simple Port-Port test	100%	40 / 40 (100%)	n/a
		1.2.1.2 Short Packet test	100%	10 / 10 (100%)	n/a
		1.2.1.3 Random test	100%	5 / 5 (100%)	n/a
		1.2.1.4 Multiport sequential_test	100%	5 / 5 (100%)	n/a
		1.2.1.5 Medium Packet Fixed	100%	10 / 10 (100%)	n/a
		1.2.1.6 Long Packet Fixed	100%	10 / 10 (100%)	n/a
		1.2.1.7 Fixed Length Fixed Delay	100%	10 / 10 (100%)	n/a
		1.2.2 Assertions_slash_Checkers for TX interface	100%	40 / 40 (100%)	100%
		1.3 RX Interface	100%	72 / 72 (100%)	100%
		1.4 Burst Mode	0%	0 / 2 (0%)	0%
		1.5 HTOC Protocol	0%	0 / 2 (0%)	0%
		1.6 Functional Coverage	40%	636 / 639 (99.53%)	0%
		1.6.1 System Interface	0%	0 / 1 (0%)	0%
		1.6.2 TX Interface	100%	628 / 628 (100%)	n/a
		1.6.3 RX Interface	100%	8 / 8 (100%)	n/a
		1.6.4 Burst Mode	0%	0 / 1 (0%)	0%
		1.6.5 HTOC Protocol	0%	0 / 1 (0%)	0%
		1.7 Code Coverage	98.69%	13444 / 13572 (99.06%)	100%

Ex	URI	Name	Overall Average Grade	Overall Covered	Assertion Status Grade
		(no filter)	(no filter)	(no filter)	(no filter)
		Overall	100%	90 / 90 (100%)	
		Code	n/a	0 / 0 (n/a)	
		Block	n/a	0 / 0 (n/a)	
		Statement	n/a	0 / 0 (n/a)	
		Expression	n/a	0 / 0 (n/a)	
		Toggle	n/a	0 / 0 (n/a)	
		FSM	n/a	0 / 0 (n/a)	
		Functional	n/a	0 / 0 (n/a)	
		Assertion	n/a	0 / 0 (n/a)	
		CoverGroup	n/a	0 / 0 (n/a)	

2.tx assertions :

Ex	URI	Name	Overall Average Grade	Overall Covered	Assertion Status Grade
		(no filter)	(no filter)	(no filter)	(no filter)
		1.2.2.1 tx ouport and vc req deassert	100%	4 / 4 (100%)	100%
		1.2.2.2 assert valid packet transfer	100%	4 / 4 (100%)	100%
		1.2.2.3 assert_tx_rel_gnt_tx_eot	100%	4 / 4 (100%)	100%
		1.2.2.4 assert_tx_eot_single_cycle	100%	4 / 4 (100%)	100%
		1.2.2.5 assert_tx_one_hot	100%	4 / 4 (100%)	100%
		1.2.2.6 assert_tx_vc_sot_vc_gnt_1	100%	4 / 4 (100%)	100%
		1.2.2.7 assert_tx_vc_sot_vc_gnt_0	100%	4 / 4 (100%)	100%
		1.2.2.8 assert_tx_vc_req_outport_req	100%	4 / 4 (100%)	100%
		1.2.2.9 assert_tx_outport_req_vc_req	100%	4 / 4 (100%)	100%
		1.2.2.10 assert_tx_req_one_hot	100%	4 / 4 (100%)	100%
		1.3 RX Interface	100%	72 / 72 (100%)	100%
		1.4 Burst Mode	0%	0 / 2 (0%)	0%
		1.5 HTOC Protocol	0%	0 / 2 (0%)	0%
		1.6 Functional Coverage	40%	636 / 639 (99.53%)	0%
		1.6.1 System Interface	0%	0 / 1 (0%)	0%
		1.6.2 TX Interface	100%	628 / 628 (100%)	n/a
		1.6.3 RX Interface	100%	8 / 8 (100%)	n/a
		1.6.4 Burst Mode	0%	0 / 1 (0%)	0%
		1.6.5 HTOC Protocol	0%	0 / 1 (0%)	0%
		1.7 Code Coverage	98.69%	13444 / 13572 (99.06%)	100%

Ex	URI	Name	Overall Average Grade	Overall Covered	Assertion Status Grade
		(no filter)	(no filter)	(no filter)	(no filter)
		Overall	100%	40 / 40 (100%)	
		Code	n/a	0 / 0 (n/a)	
		Block	n/a	0 / 0 (n/a)	
		Statement	n/a	0 / 0 (n/a)	
		Expression	n/a	0 / 0 (n/a)	
		Toggle	n/a	0 / 0 (n/a)	
		FSM	n/a	0 / 0 (n/a)	
		Functional	100%	40 / 40 (100%)	

3.rx interface tests:

Ex	URI	Name	Overall Average Grade	Overall Covered	Assertion Status Grade
		(no filter)	(no filter)	(no filter)	(no filter)
		1.3.1.1 Simple_port-port test	100%	10 / 10 (100%)	n/a
		1.3.1.2 Short Packet Test	100%	10 / 10 (100%)	n/a
		1.3.1.3 Random_Test	100%	5 / 5 (100%)	n/a
		1.3.1.4 Multiport sequential Test	100%	5 / 5 (100%)	n/a
		1.3.1.5 Medium Packet Fixed	100%	10 / 10 (100%)	n/a
		1.3.1.6 Long Packet Fixed	100%	10 / 10 (100%)	n/a
		1.3.1.7 Fixed Length Delay	100%	10 / 10 (100%)	n/a
		1.3.2 Assertions_slash_Checkers for RX interface	100%	12 / 12 (100%)	100%
		1.4 Burst Mode	0%	0 / 2 (0%)	0%
		1.5 HTOC Protocol	0%	0 / 2 (0%)	0%
		1.6 Functional Coverage	40%	636 / 639 (99.53%)	0%
		1.6.1 System Interface	0%	0 / 1 (0%)	0%
		1.6.2 TX Interface	100%	628 / 628 (100%)	n/a
		1.6.3 RX Interface	100%	8 / 8 (100%)	n/a
		1.6.4 Burst Mode	0%	0 / 1 (0%)	0%
		1.6.5 HTOC Protocol	0%	0 / 1 (0%)	0%
		1.7 Code Coverage	98.69%	13444 / 13572 (99.06%)	100%

Ex	URI	Name	Overall Average Grade	Overall Covered	Assertion Status Grade
		(no filter)	(no filter)	(no filter)	(no filter)
		Overall	100%	60 / 60 (100%)	
		Code	n/a	0 / 0 (n/a)	
		Block	n/a	0 / 0 (n/a)	
		Statement	n/a	0 / 0 (n/a)	
		Expression	n/a	0 / 0 (n/a)	
		Toggle	n/a	0 / 0 (n/a)	
		FSM	n/a	0 / 0 (n/a)	
		Functional	n/a	0 / 0 (n/a)	
		Assertion	n/a	0 / 0 (n/a)	
		CoverGroup	n/a	0 / 0 (n/a)	

4.rx assertions:

ICE 616 Project Fall2024

Uti

Name

Overall Average Grade

Overall Covered

Assertion Status Grade

(no filter)

(no filter)

(no filter)

(no filter)

CSCE 616 Project Fall2024

48.38%

14282 / 14419 (99.05%)

96.65%

1 HTAX_v014

48.38%

14282 / 14419 (99.05%)

96.65%

1.1 System Interface

0%

0 / 2 (0%)

0%

1.2 TX Interface

100%

130 / 130 (100%)

100%

1.2.1 Testcases to verify TX interface

100%

90 / 90 (100%)

n/a

1.2.2 Assertions_slash_Checkers for TX interface

100%

40 / 40 (100%)

100%

1.3 RX Interface

100%

72 / 72 (100%)

100%

1.3.1 Testcases to verify RX interface

100%

60 / 60 (100%)

n/a

1.3.2 Assertions_slash_Checkers for RX interface

100%

12 / 12 (100%)

100%

1.3.2.1 assert_rx_eot_one_cycle

100%

4 / 4 (100%)

100%

1.3.2.2 assert_rx_sot_one_hot

100%

4 / 4 (100%)

100%

1.3.2.3 assert_eot_timeout_check

100%

4 / 4 (100%)

100%

1.4 Burst Mode

0%

0 / 2 (0%)

0%

1.5 HTOC Protocol

0%

0 / 2 (0%)

0%

1.6 Functional Coverage

40%

636 / 639 (99.53%)

0%

1.6.1 System Interface

0%

0 / 1 (0%)

0%

1.6.2 TX Interface

100%

628 / 628 (100%)

n/a

1.6.3 RX Interface

100%

8 / 8 (100%)

n/a

1.6.4 Burst Mode

0%

0 / 1 (0%)

0%

1.6.5 HTOC Protocol

0%

0 / 1 (0%)

0%

1.7 Code Coverage

98.69%

13444 / 13572 (99.06%)

100%

Recursive

Uti

Name

Overall Average Grade

Overall Covered

Assertion Status Grade

(no filter)

(no filter)

(no filter)

(no filter)

1.3.2.1 assert_rx_eot_one_cycle

100%

4 / 4 (100%)

100%

1.3.2.2 assert_rx_sot_one_hot

100%

4 / 4 (100%)

100%

1.3.2.3 assert_eot_timeout_check

100%

4 / 4 (100%)

100%

Showing 3 items

Details

Assertions_slash_Checkers for RX interface

Metrics

Source

Attributes

Uti

Name

Overall Average Grade

Overall Covered

Overall

100%

12 / 12 (100%)

Code

n/a

0 / 0 (n/a)

Block

n/a

0 / 0 (n/a)

Statement

n/a

0 / 0 (n/a)

Expression

n/a

0 / 0 (n/a)

Toggle

n/a

0 / 0 (n/a)

5.tx functional coverage:

CSCE 616 Project Fall2024

Uti

Name

Overall Average Grade

Overall Covered

Assertion Status Grade

(no filter)

(no filter)

(no filter)

(no filter)

CSCE 616 Project Fall2024

48.38%

14282 / 14419 (99.05%)

96.65%

1 HTAX_v014

48.38%

14282 / 14419 (99.05%)

96.65%

1.1 System Interface

0%

0 / 2 (0%)

0%

1.2 TX Interface

100%

130 / 130 (100%)

100%

1.3 RX Interface

100%

72 / 72 (100%)

100%

1.4 Burst Mode

0%

0 / 2 (0%)

0%

1.5 HTOC Protocol

0%

0 / 2 (0%)

0%

1.6 Functional Coverage

40%

636 / 639 (99.53%)

0%

1.6.1 System Interface

0%

0 / 1 (0%)

0%

1.6.2 TX Interface

100%

628 / 628 (100%)

n/a

1.6.2.1 VC Request

100%

12 / 12 (100%)

n/a

1.6.2.2 Outport Request

100%

16 / 16 (100%)

n/a

1.6.2.3 TX_VC_GNT

100%

12 / 12 (100%)

n/a

1.6.2.4 DEST_PORT

100%

16 / 16 (100%)

n/a

1.6.2.5 VC

100%

12 / 12 (100%)

n/a

1.6.2.6 LENGTH

100%

64 / 64 (100%)

n/a

1.6.2.7 DEST_PORT_VC

100%

48 / 48 (100%)

n/a

1.6.2.8 DEST_PORT_LENGTH

100%

256 / 256 (100%)

n/a

1.6.2.9 X_VC_LENGTH

100%

192 / 192 (100%)

n/a

1.6.3 RX Interface

100%

8 / 8 (100%)

n/a

1.6.4 Burst Mode

0%

0 / 1 (0%)

0%

1.6.5 HTOC Protocol

0%

0 / 1 (0%)

0%

1.7 Code Coverage

98.69%

13444 / 13572 (99.06%)

100%

Recursive

Uti

Name

Overall Average Grade

Overall Covered

Assertion Status Grade

(no filter)

(no filter)

(no filter)

(no filter)

1.6.2.1 VC Request

100%

12 / 12 (100%)

n/a

1.6.2.2 Outport Request

100%

16 / 16 (100%)

n/a

1.6.2.3 TX_VC_GNT

100%

12 / 12 (100%)

n/a

1.6.2.4 DEST_PORT

100%

16 / 16 (100%)

n/a

1.6.2.5 VC

100%

12 / 12 (100%)

n/a

1.6.2.6 LENGTH

100%

64 / 64 (100%)

n/a

1.6.2.7 DEST_PORT_VC

100%

48 / 48 (100%)

n/a

1.6.2.8 DEST_PORT_LENGTH

100%

256 / 256 (100%)

n/a

1.6.2.9 X_VC_LENGTH

100%

192 / 192 (100%)

n/a

Showing 9 items

Details

TX Interface

Metrics

Source

Attributes

Uti

Name

Overall Average Grade

Overall Covered

1 Overall

100%

628 / 628 (100%)

1 Code

n/a

0 / 0 (n/a)

1 Block

n/a

0 / 0 (n/a)

1 Statement

n/a

0 / 0 (n/a)

1 Expression

n/a

0 / 0 (n/a)

1 Toggle

n/a

0 / 0 (n/a)

1 FSM

n/a

0 / 0 (n/a)

1 Functional

100%

628 / 628 (100%)

6.rx functional coverage:

vPlan Hierarchy			
Name	Overall Average Grade	Overall Covered	Assertion Status Grade
(no filter)	(no filter)	(no filter)	(no filter)
CSCE 616 Project Fall2024	48.38%	14282 / 14419 (99.05%)	96.65%
1 HTAX_v014	48.38%	14282 / 14419 (99.05%)	96.65%
1.1 System Interface	0%	0 / 2 (0%)	0%
1.2 TX Interface	100%	130 / 130 (100%)	100%
1.3 RX Interface	100%	72 / 72 (100%)	100%
1.4 Burst Mode	0%	0 / 2 (0%)	0%
1.5 HTOC Protocol	0%	0 / 2 (0%)	0%
1.6 Functional Coverage	40%	636 / 639 (99.53%)	0%
1.6.1 System Interface	0%	0 / 1 (0%)	0%
1.6.2 TX Interface	100%	628 / 628 (100%)	n/a
1.6.3 RX Interface	100%	8 / 8 (100%)	n/a
1.6.3.1 RX DATA	100%	4 / 4 (100%)	n/a
1.6.3.2 RX EOT	100%	4 / 4 (100%)	n/a
1.6.4 Burst Mode	0%	0 / 1 (0%)	0%
1.6.5 HTOC Protocol	0%	0 / 1 (0%)	0%
1.7 Code Coverage	98.69%	13444 / 13572 (99.06%)	100%

Recursive			
URI Name	Overall Average Grade	Overall Covered	Assertion Status Grade
(no filter)	(no filter)	(no filter)	(no filter)
1.6.3.1 RX DATA	100%	4 / 4 (100%)	n/a
1.6.3.2 RX EOT	100%	4 / 4 (100%)	n/a

Metrics			
URI Name	Source	Attributes	
Overall			Overall Average Grade: 100% Overall Covered: 8 / 8 (100%)
Code			0 / 0 (n/a)
Block			0 / 0 (n/a)
Statement			0 / 0 (n/a)
Expression			0 / 0 (n/a)
Toggle			0 / 0 (n/a)

7.Code coverage:

vPlan Hierarchy			
Name	Overall Average Grade	Overall Covered	Assertion Status Grade
(no filter)	(no filter)	(no filter)	(no filter)
CSCE 616 Project Fall2024	48.38%	14282 / 14419 (99.05%)	96.65%
1 HTAX_v014	48.38%	14282 / 14419 (99.05%)	96.65%
1.1 System Interface	0%	0 / 2 (0%)	0%
1.2 TX Interface	100%	130 / 130 (100%)	100%
1.3 RX Interface	100%	72 / 72 (100%)	100%
1.4 Burst Mode	0%	0 / 2 (0%)	0%
1.5 HTOC Protocol	0%	0 / 2 (0%)	0%
1.6 Functional Coverage	40%	636 / 639 (99.53%)	0%
1.6.1 System Interface	0%	0 / 1 (0%)	0%
1.6.2 TX Interface	100%	628 / 628 (100%)	n/a
1.6.3 RX Interface	100%	8 / 8 (100%)	n/a
1.6.4 Burst Mode	0%	0 / 1 (0%)	0%
1.6.5 HTOC Protocol	0%	0 / 1 (0%)	0%
1.7 Code Coverage	98.69%	13444 / 13572 (99.06%)	100%
1.7.1 Block	98.69%	3361 / 3393 (99.06%)	100%
1.7.2 Expression	98.69%	3361 / 3393 (99.06%)	100%
1.7.3 Toggle	98.69%	3361 / 3393 (99.06%)	100%
1.7.4 FSM	98.69%	3361 / 3393 (99.06%)	100%

Recursive			
URI Name	Overall Average Grade	Overall Covered	Assertion Status Grade
(no filter)	(no filter)	(no filter)	(no filter)
1.7.1 Block	98.69%	3361 / 3393	100%
1.7.2 Expression	98.69%	3361 / 3393	100%
1.7.3 Toggle	98.69%	3361 / 3393	100%
1.7.4 FSM	98.69%	3361 / 3393	100%

Metrics			
URI Name	Source	Attributes	
Overall			Overall Average Grade: 98.69% Overall Covered: 13444 / 13572 (99.06%)
Code			98.69% 13236 / 13364 (99.06%)
Block			96.41% 852 / 884 (96.38%)
Statement			n/a 0 / 0 (n/a)
Expression			93.21% 352 / 384 (91.67%)
Toggle			99.89% 12032 / 12096 (99.48%)
FSM			n/a 0 / 0 (n/a)

Bug Report:

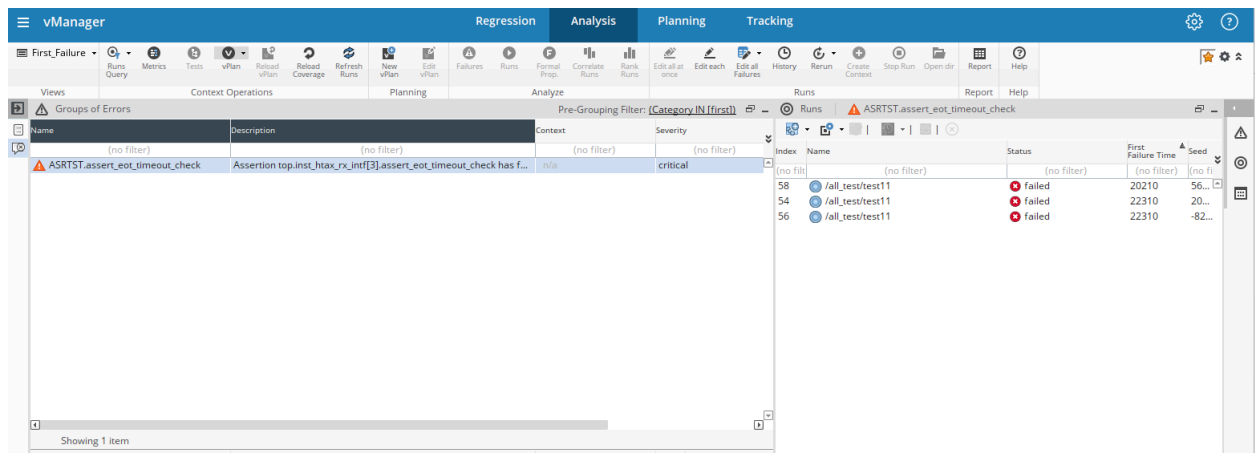
Design file name:htax_outport_data_mux

Line number:43

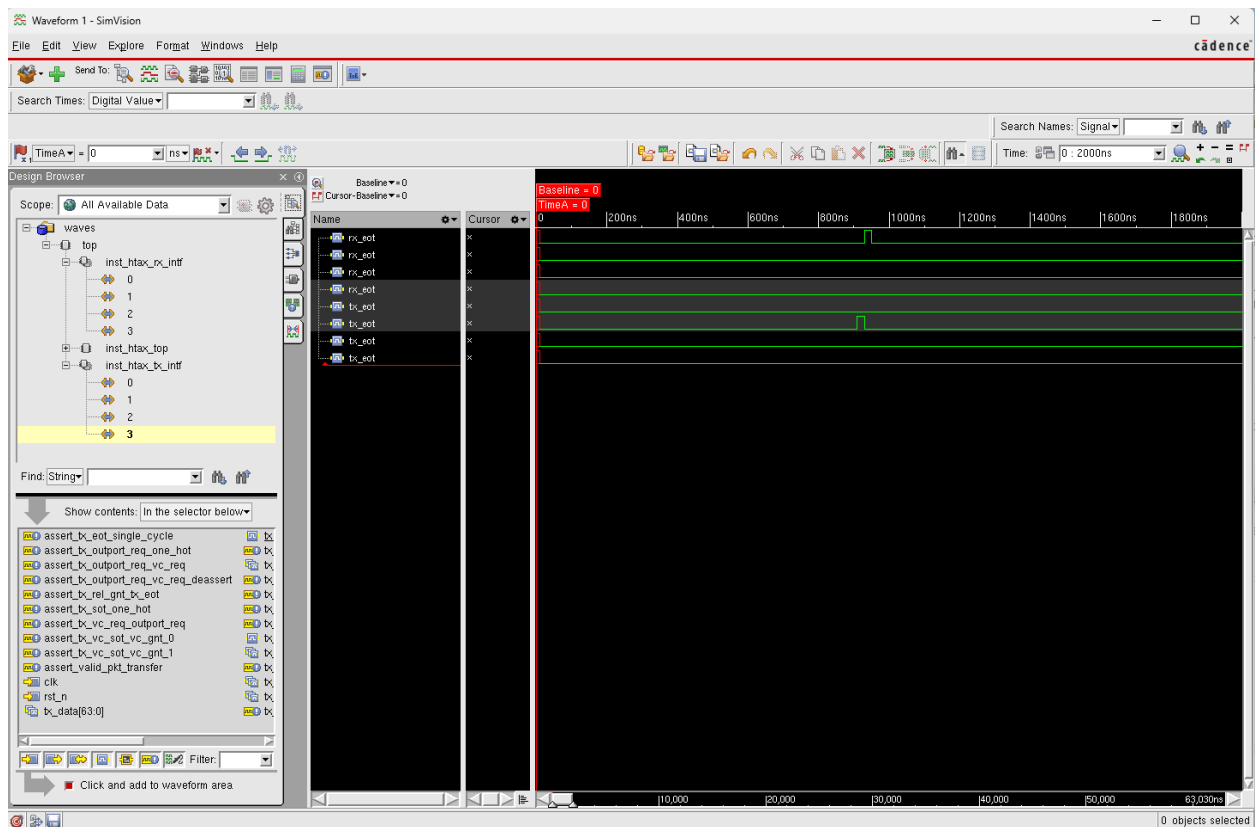
Wrong code:|(eot_in & inport_sel_reg) & ~(&(eot_in));

Corrected one:assign selected_eot = |(eot_in & inport_sel_reg);

Failed Analysis due to bug:



Waveform before the bug fix:(no rising eot)



Waveform after bug fixing:(the rx eot)

